

Variational Quantum Eigensolver (VQE) for the H₂ Molecule

Dissociation Curve, Exact Baseline, and a Path Toward Drug-Target Binding Energies

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1. Problem Statement

1.1 The scientific problem

Drug discovery depends on predicting how strongly a small molecule (ligand) binds to a biological target (protein binding site). The key quantity is the **binding energy**:

$$\Delta E_{\text{bind}} = E_{\text{complex}} - E_{\text{ligand}} - E_{\text{target}}$$

Each of these energies is the ground-state energy of a many-electron quantum system. Computing it exactly requires solving the electronic Schrödinger equation, whose cost on a classical computer grows **exponentially** with the number of electrons and orbitals. Approximate classical methods (DFT, coupled cluster) trade accuracy for speed, and they struggle with strongly correlated systems.

1.2 The quantum-computing opportunity

Quantum computers represent many-electron wavefunctions naturally: n qubits hold a state in a 2^n -dimensional space. The **Variational Quantum Eigensolver (VQE)** is a hybrid quantum-classical algorithm suited to today's noisy, small devices (NISQ era) for estimating ground-state energies.

1.3 Why the original prototype was not enough

The first version of this project (`QDrugTargetPipeline`) had these weaknesses:

Issue	Consequence
Hamiltonian was hard-coded (<code>(ZI, IZ, XX, ZZ)</code> with made-up coefficients)	The result was not tied to any real molecule
Atomic coordinates only produced one number, <code>(J = 1/d²)</code>	Geometry barely influenced the quantum calculation
The "ground-state" line in the plot was <code>(min(energy_history))</code>	Circular: it did not compare against a true reference
Random initial parameters, single run	Not reproducible; COBYLA can get stuck in local minima
Units (Hartree, Å) were stated but the Hamiltonian was dimensionless	Labels overstated what was computed

1.4 Problem solved in this document

Compute the ground-state energy of the H₂ molecule as a function of the H-H bond distance using VQE, validate it against an exact classical solution, and report the error relative to chemical accuracy.

H₂ is the standard "hello world" of quantum chemistry. It is small enough to verify by hand (Section 4), and it exercises the full pipeline (real Hamiltonian, qubit mapping, ansatz, optimizer, validation) that a larger drug-target calculation would reuse.

2. Objectives

1. Generate a **real molecular Hamiltonian** for H₂ from first principles (PySCF, STO-3G basis).
2. Map the fermionic Hamiltonian to a **2-qubit** operator (parity mapping with two-qubit reduction).
3. Build a **UCCSD** ansatz starting from the **Hartree-Fock** reference state.
4. Minimize the energy with a classical optimizer (**COBYLA**) using exact statevector simulation.
5. Compare against **exact diagonalization** at every geometry.
6. Plot the **dissociation curve** and the **error versus chemical accuracy (1.6 mHa)**.

3. Theory

3.1 The electronic structure problem

Born-Oppenheimer approximation

Nuclei are thousands of times heavier than electrons, so we treat them as fixed point charges and solve only for the electrons. For fixed nuclear positions $\mathbf{R}$, the electronic Hamiltonian (atomic units) is:

$$\hat{H}_{\text{el}} = - \sum_i \frac{1}{2} \nabla_i^2 - \sum_{i,A} \frac{Z_A}{|\mathbf{r}_i - \mathbf{R}_A|} + \sum_{i < j} \frac{1}{|\mathbf{r}_i - \mathbf{r}_j|}$$

The three terms are the electron kinetic energy, electron-nucleus attraction, and electron-electron repulsion. The **total energy** adds the classical nuclear repulsion:

$$E_{\text{total}}(\mathbf{R}) = E_{\text{el}}(\mathbf{R}) + E_{\text{nuc}}(\mathbf{R}), \quad E_{\text{nuc}} = \sum_{A < B} \frac{Z_A Z_B}{|\mathbf{R}_A - \mathbf{R}_B|}$$

Scanning $\mathbf{R}$ gives the **potential energy surface** (for H_2 , a 1-D curve).

Units

- Energy: **Hartree** (Ha). $1 \text{ Ha} \approx 627.5 \text{ kcal/mol} \approx 27.211 \text{ eV}$.
- Length in the formulas: **Bohr** ($a_0 = 0.529177 \text{ \AA}$). The code takes $\text{\AA}$ and PySCF converts internally.

3.2 Basis set: STO-3G

Molecular orbitals are built from atomic basis functions. **STO-3G** approximates each Slater-type orbital with 3 Gaussians. For H_2 it gives 1 basis function per hydrogen, so:

- 2 spatial orbitals $\rightarrow$ **4 spin orbitals**
- 2 electrons

This is a *minimal* basis. It is qualitatively right and exactly solvable, but it is not quantitatively accurate against experiment.

3.3 Hartree-Fock and electron correlation

Hartree-Fock (HF) approximates the wavefunction as a single Slater determinant: both electrons in the bonding orbital σ_g . The energy missing relative to the exact answer (within the same basis) is the **correlation energy**:

$$E_{\text{corr}} = E_{\text{FCI}} - E_{\text{HF}}$$

For H_2 near equilibrium, $E_{\text{corr}} \approx -0.02 \text{ Ha}$. Recovering it is the whole point of VQE here. HF fails badly at large bond lengths, where the true state mixes the bonding configuration $(\sigma_g)^2$ with the anti-bonding one $(\sigma_u)^2$.

FCI (Full Configuration Interaction) is the exact solution within the chosen basis. For $\text{H}_2/\text{STO-3G}$ the FCI space is tiny (it can be diagonalized exactly), so it is our reference.

3.4 Second quantization

The Hamiltonian is written with fermionic creation/annihilation operators:

$$\hat{H} = \sum_{pq} h_{pq} a_p^\dagger a_q + \frac{1}{2} \sum_{pqrs} h_{pqrs} a_p^\dagger a_q^\dagger a_r a_s$$

- `h_pq`: one-electron integrals (kinetic + nuclear attraction)
- `h_pqrs`: two-electron integrals (electron repulsion)

PySCF computes these integrals; `qiskit-nature` assembles the operator.

3.5 Fermion-to-qubit mapping

Qubits are distinguishable, while electrons are fermions, so the operators must be mapped while preserving anticommutation.

Mapping	Qubits for H ₂ /STO-3G	Notes
Jordan-Wigner	4	One qubit per spin orbital; simple but non-local Z strings
Bravyi-Kitaev	4	Balanced locality
Parity	4 → 2	Allows two-qubit reduction using particle-number and spin symmetry

We use **parity mapping with two-qubit reduction**. The resulting Hamiltonian is a sum of Pauli strings:

$$\hat{H}_{\text{qubit}} = \sum_k c_k P_k, \quad P_k \in \{I, X, Y, Z\}^{\otimes n}$$

Fewer qubits mean shallower circuits, which matters a lot on noisy hardware.

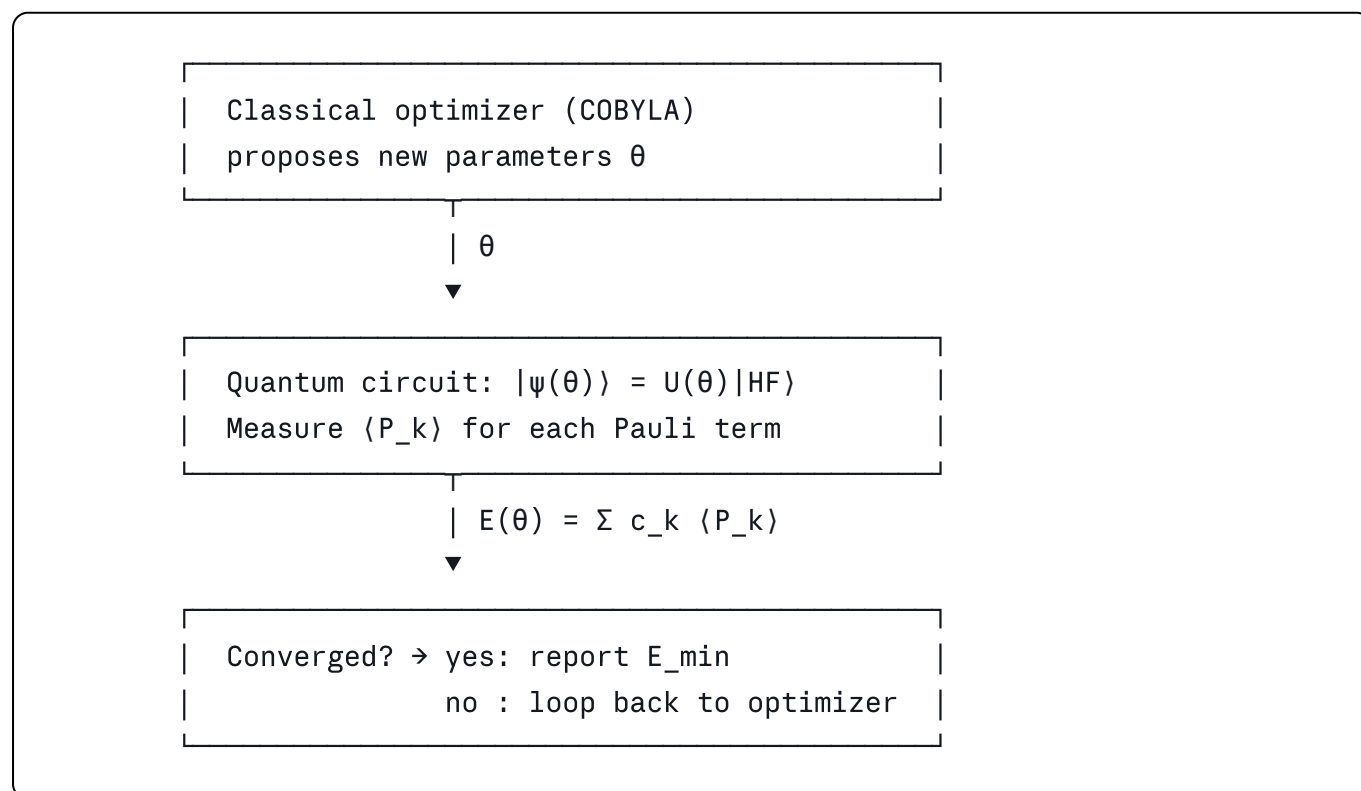
3.6 The variational principle

For any normalized trial state $|\psi(\theta)\rangle$:

$$E(\boldsymbol{\theta}) = \langle \psi(\boldsymbol{\theta}) | \hat{H} | \psi(\boldsymbol{\theta}) \rangle \geq E_0$$

where E_0 is the true ground-state energy. This is the foundation of VQE. The energy of any circuit you build is an **upper bound** on the truth, so minimizing over θ moves you toward E_0 from above. **A VQE energy below the exact energy indicates a bug** (wrong sector, missing nuclear repulsion, wrong geometry, and so on).

3.7 The VQE algorithm



1. Prepare the reference state $|\text{HF}\rangle$ (qubits flipped to occupy the lowest orbitals).
2. Apply a parameterized unitary $U(\theta)$, the **ansatz**.
3. Estimate $E(\theta) = \sum_k c_k \langle P_k \rangle$ (here exactly, via statevector; on hardware, by sampling).
4. A classical optimizer updates θ ; repeat until convergence.
5. Add the nuclear repulsion to get the total energy.

3.8 The UCCSD ansatz

Unitary Coupled Cluster with Singles and Doubles is chemically motivated:

$$|\psi(\theta)\rangle = e^{\hat{T}(\theta) - \hat{T}^\dagger(\theta)} |\text{HF}\rangle, \quad \hat{T} = \hat{T}_1 + \hat{T}_2$$

- $\hat{T}_1$ (singles): moves one electron from an occupied to a virtual orbital.
- $\hat{T}_2$ (doubles): moves two electrons at once.

For $\text{H}_2/\text{STO-3G}$ the physically important excitation is the **double excitation** $(\sigma_g)^2 \rightarrow (\sigma_u)^2$. Within the reduced 2-qubit space the ansatz effectively rotates between the Hartree-Fock configuration and the doubly excited one by a single angle (see Section 4.4).

Why start from Hartree-Fock? With $\theta = 0$, UCCSD reproduces HF exactly, so the optimizer begins at a physically sensible point and only has to find the *correlation correction*. Random starts often land in worse local minima.

3.9 The classical optimizer: COBYLA

COBYLA (Constrained Optimization BY Linear Approximation) is **gradient-free**. It only needs function values, so it works with noisy estimates and with any ansatz. Its weakness is that it can stall in local minima and may need many evaluations as parameters grow. This is why the script supports **multiple restarts**.

3.10 Chemical accuracy

Chemists consider a method useful for energy differences if it reaches **1 kcal/mol**:

$$1 \text{ kcal/mol} = 1.594 \times 10^{-3} \text{ Ha} \approx 1.6 \text{ mHa}$$

The lower plot in the script draws this threshold as a dashed line.

4. Worked Calculation Example (H₂ at 0.735 Å)

This section shows by hand what the code does internally. Values are those typically obtained with PySCF/STO-3G; the last digit or sign convention can differ slightly between library versions.

4.1 Step 1: Nuclear repulsion energy

Bond length: 0.735 Å = 0.735 / 0.529177 = **1.3889 Bohr**.

$$E_{\text{nuc}} = \frac{Z_A Z_B}{R} = \frac{1 \times 1}{1.3889} \approx \mathbf{0.7200 \text{ Ha}}$$

4.2 Step 2: Qubit Hamiltonian

After computing integrals, building the second-quantized operator, and applying parity mapping with two-qubit reduction, the **electronic** Hamiltonian is:

$$\hat{H} = c_0 II + c_1 IZ + c_2 ZI + c_3 ZZ + c_4 XX$$

Pauli term	Coefficient (Ha)
II	-1.05237
IZ	+0.39794
ZI	-0.39794
ZZ	-0.01128
XX	+0.18093

(Qiskit uses little-endian ordering: the rightmost character acts on qubit 0.)

4.3 Step 3: Build the 4×4 matrix and diagonalize by hand

Basis states $|q_1 q_0\rangle$. The Z-type terms are diagonal. Their eigenvalues are:

State	IZ	ZI	ZZ
$ 00\rangle$	+1	+1	+1
$ 01\rangle$	-1	+1	-1
$ 10\rangle$	+1	-1	-1
$ 11\rangle$	-1	-1	+1

Diagonal entries, $H_{ii} = c_0 + c_1 \cdot (IZ) + c_2 \cdot (ZI) + c_3 \cdot (ZZ)$:

State	Calculation	H_{ii} (Ha)
$ 00\rangle$	$-1.05237 + 0.39794 - 0.39794 - 0.01128$	-1.06365
$ 01\rangle$	$-1.05237 - 0.39794 - 0.39794 + 0.01128$	-1.83697
$ 10\rangle$	$-1.05237 + 0.39794 + 0.39794 + 0.01128$	-0.24523
$ 11\rangle$	$-1.05237 - 0.39794 + 0.39794 - 0.01128$	-1.06365

The **XX** term flips both qubits, so it couples only $|00\rangle \leftrightarrow |11\rangle$ and $|01\rangle \leftrightarrow |10\rangle$ with strength $c_4 = 0.18093$. The matrix therefore splits into two independent 2×2 blocks.

Interpretation: $|01\rangle$ is the **Hartree-Fock state** (energy -1.83697 Ha, electronic part). The other block corresponds to different particle-number or spin sectors.

Block A: $\{|01\rangle, |10\rangle\}$ (the physically relevant sector)

$$H_A = \begin{pmatrix} -1.83697 & 0.18093 \\ 0.18093 & -0.24523 \end{pmatrix}$$

Eigenvalues of a 2×2 symmetric matrix:

$$\lambda_{\pm} = \frac{H_{11} + H_{22}}{2} \pm \sqrt{\left(\frac{H_{11} - H_{22}}{2}\right)^2 + H_{12}^2}$$

- Mean: $(-1.83697 - 0.24523)/2 = \mathbf{-1.04110}$
- Half-difference: $(-1.83697 + 0.24523)/2 = \mathbf{-0.79587}$
- Square root: $\sqrt{(0.79587^2 + 0.18093^2)} = \sqrt{(0.63341 + 0.03274)} = \sqrt{0.66615} = \mathbf{0.81618}$

$$\lambda_- = -1.04110 - 0.81618 = \mathbf{-1.85728 \text{ Ha}} \quad (\text{electronic ground state})$$

$$\lambda_+ = -1.04110 + 0.81618 = -0.22492 \text{ Ha}$$

Block B: $\{|00\rangle, |11\rangle\}$

$$H_B = \begin{pmatrix} -1.06365 & 0.18093 \\ 0.18093 & -1.06365 \end{pmatrix} \Rightarrow \lambda = -1.06365 \pm 0.18093 = -1.24458, -0.88272$$

These sit higher than -1.85728 Ha , so Block A contains the true ground state. (This is also why the script obtains its reference through `GroundStateEigensolver`, which respects the correct particle-number sector, instead of naively taking the lowest eigenvalue of an arbitrary matrix.)

4.4 Step 4: What VQE does in this 2×2 block

Within Block A, the UCCSD ansatz acting on $|HF\rangle = |01\rangle$ produces a state of the form:

$$|\psi(\theta)\rangle = \cos \theta |01\rangle + \sin \theta |10\rangle$$

(The sign of θ depends on the library's convention.) Its energy:

$$E(\theta) = H_{11} \cos^2 \theta + H_{22} \sin^2 \theta + 2H_{12} \sin \theta \cos \theta$$

- **At $\theta = 0$ (pure Hartree-Fock):** $E = H_{11} = -1.83697 \text{ Ha}$
- **Optimum** satisfies $\tan 2\theta = 2H_{12} / (H_{11} - H_{22}) = 0.36186 / (-1.59174) = -0.2273 \rightarrow 2\theta \approx -0.2235 \text{ rad} \rightarrow |\theta| \approx \mathbf{0.112 \text{ rad}}$
- **At the optimum:** $E = \lambda_- = -1.85728 \text{ Ha}$

The small rotation angle shows that the ground state is mostly the HF configuration with a small admixture of the doubly excited one. That admixture is the correlation energy.

4.5 Step 5: Total energy and correlation energy

Quantity	Electronic (Ha)	+ E_nuc (0.7200) = Total (Ha)
Hartree-Fock ($\theta = 0$)	-1.83697	-1.1170
VQE / FCI (θ optimal)	-1.85728	-1.1373

$$E_{\text{corr}} = -1.85728 - (-1.83697) = -\mathbf{0.0203 \text{ Ha}} (\approx -12.7 \text{ kcal/mol})$$

The correlation energy of about 20 mHa is **more than 12× larger than chemical accuracy**. A method that stops at Hartree-Fock is not chemically accurate, while VQE with UCCSD recovers it.

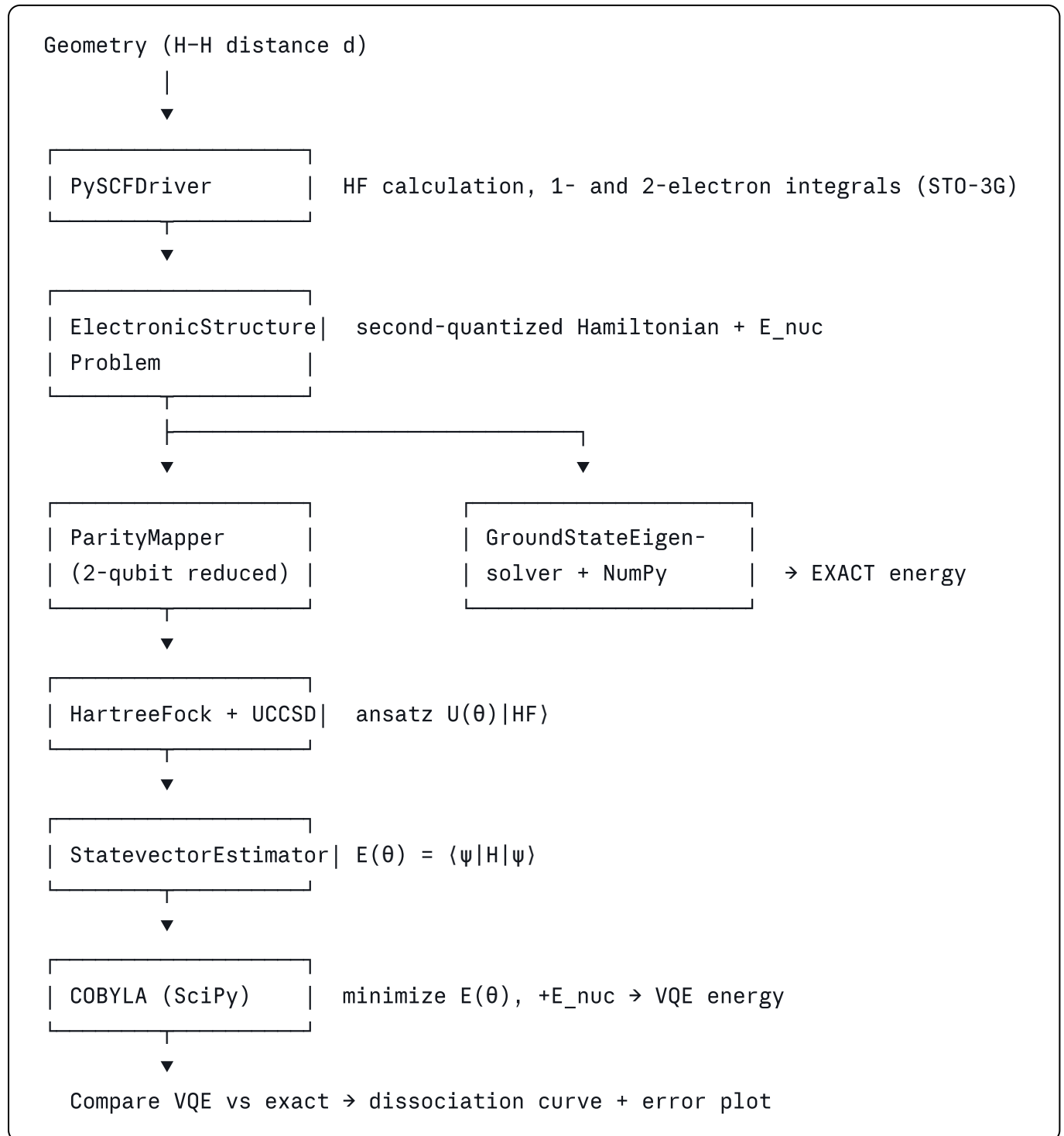
4.6 Step 6: What the script prints

For $d = 0.74 \text{ \AA}$ the output line looks like:

d=0.74 Å | exact=-1.137306 | VQE=-1.137306 | err=1.2e-07 Ha

(The exact digits depend on the scan grid, which uses `np.linspace(0.4, 2.5, 15)` and so does not land exactly on 0.735 Å.)

5. System Architecture and Workflow



6. Code Walkthrough

File: `h2_vqe_scan.py`

6.1 `build_problem(distance)`

Creates a PySCF driver for two hydrogen atoms separated along the z-axis:

```
python
```

```
atom=f"H 0 0 0; H 0 0 {distance}", basis="sto3g", charge=0, spin=0
```

- `charge=0, spin=0`: neutral, closed-shell (2 electrons, spin singlet).
- `driver.run()` returns an `ElectronicStructureProblem` holding the integrals, the number of particles, the number of spatial orbitals, and the nuclear repulsion energy.

6.2 `exact_energy(problem, mapper)`

```
python
```

```
solver = GroundStateEigensolver(mapper, NumPyMinimumEigensolver())
```

Uses classical exact diagonalization of the mapped Hamiltonian within the correct symmetry sector. It returns the **total** energy (electronic + nuclear). This is the reference for validation.

6.3 `vqe_energy(problem, mapper, seed, restarts)`

Step	Code	Purpose
Map	<code>mapper.map(problem.hamiltonian.second_q_op())</code>	Fermionic $\rightarrow$ Pauli operator (2 qubits)
Nuclear term	<code>problem.hamiltonian.nuclear_repulsion_energy</code>	Added back because the qubit operator contains only electronic energy
Reference state	<code>HartreeFock(...)</code>	Initial occupation
Ansatz	<code>UCCSD(..., initial_state=hf)</code>	Chemically motivated $U(\theta)$
Cost	<code>estimator.run([(ansatz, hamiltonian, params)])</code>	Exact $\langle H \rangle$ via statevector
Optimize	<code>minimize(cost, x0, method="COBYLA", maxiter=300)</code>	Classical loop
Restarts	restart 0 starts at $\theta = 0$ (= HF); others use small seeded random values	Escape local minima, stay reproducible

It returns the best total energy found and its convergence history.

6.4 `main()`

1. Scans 15 distances from 0.4 to 2.5 Å.
2. At each distance: builds the problem, computes exact and VQE energies, prints the error.
3. Produces a two-panel figure saved as `h2_dissociation.png`:
 - **Top:** energy vs distance, exact (line) vs VQE (points).
 - **Bottom:** $|VQE - exact|$ on a log scale with the chemical-accuracy threshold.
4. Prints the scan minimum (the equilibrium bond length estimate).

7. Installation and Usage

7.1 Requirements

- Python 3.9 to 3.12 recommended

- Packages: `qiskit`, `qiskit-nature` (0.7.x), `qiskit-algorithms`, `pyscf`, `scipy`, `matplotlib`

7.2 Setup (recommended: virtual environment)

```
bash

python3 -m venv ~/qenv
source ~/qenv/bin/activate
pip install qiskit qiskit-nature qiskit-algorithms pyscf scipy matplotlib
```

7.3 Run

```
bash

python -u h2_vqe_scan.py
```

7.4 Outputs

- Console: one line per distance with exact energy, VQE energy, and error.
- `h2_dissociation.png`: dissociation curve and error plot (300 dpi).

8. Expected Results and How to Read Them

8.1 The dissociation curve (top panel)

Region	Expected behavior
Short distance (≈ 0.4 Å)	Energy rises steeply (nuclear repulsion dominates)
≈ 0.74 Å	Minimum , about -1.137 Ha (experimental bond length ≈ 0.741 Å)
Large distance (≥ 2.5 Å)	Curve flattens toward ≈ -0.93 Ha, which is two separated H atoms in STO-3G (2×-0.4666 Ha)

The depth of the well (asymptote minus minimum, ≈ 0.20 Ha) is the STO-3G dissociation energy. It is larger than the experimental ≈ 0.17 Ha because the minimal basis is imperfect.

8.2 The error plot (bottom panel)

- VQE points should lie **at or below the dashed 1.6 mHa line**, and typically many orders of magnitude lower, since the ansatz is expressive enough for H_2 and the simulation is noiseless.
- Errors are **always ≥ 0** (variational principle). A negative error means a bug.

- If some distances show larger errors, COBYLA likely got stuck. Increase `restarts` or `maxiter`.

8.3 Sanity checklist

- ☐ Exact energy minimum near 0.74 Å
- ☐ Exact energy ≈ -1.137 Ha at the minimum
- ☐ VQE $\geq$ exact at every distance
- ☐ Error < 1.6 mHa everywhere

9. Limitations

Limitation	Effect	Remedy
Minimal basis (STO-3G)	Quantitative error vs experiment	Larger basis (6-31G, cc-pVDZ); more qubits
Noiseless statevector simulation	Ignores hardware noise and shot noise	Aer noise models, shot-based estimator, error mitigation (ZNE)
Tiny system (2 qubits)	Does not test scalability	LiH, H ₂ O, then active-space fragments
Gradient-free COBYLA	Slow for many parameters	SPSA, L-BFGS-B with parameter-shift gradients, ADAPT-VQE
UCCSD depth	Deep circuits for larger molecules	Hardware-efficient or ADAPT ansätze
Ground state only	No excited states or reactivity	SSVQE, qEOM

On the "quantum advantage" question: for H₂ a classical computer solves the problem instantly and exactly. This project demonstrates and validates the *method*. It does not demonstrate an advantage.

10. Extending Toward Drug-Target Binding

H₂ is the validation case. A drug-target workflow reuses the same structure:

Stage	H ₂ (this project)	Drug-target extension
Geometry	One distance parameter	Ligand + binding-site fragment from PDB/SDF (RDKit, Biopython)
Classical preprocessing	None	Docking (AutoDock Vina), force-field pose refinement (RDKit MMFF, OpenMM)
Hamiltonian	Full molecule, STO-3G	Active space around the binding site (qiskit-nature <code>ActiveSpaceTransformer</code>)
Quantity	E(d)	$\Delta E_{\text{bind}} = E(\text{complex}) - E(\text{ligand}) - E(\text{target})$
Scan	Bond distance	Pose or ligand-target distance
Validation	Exact diagonalization	Classical CASCI/CCSD(T) on the same active space
Hardware realism	Statevector	Noise models, error mitigation

Key caveats for the drug-target version:

1. A full protein cannot be simulated, so only a chemically relevant **fragment** (tens of atoms at most, a handful of active orbitals) is feasible.
2. Binding energies involve **small differences of large numbers**, so chemical accuracy matters and basis-set superposition error must be handled.
3. Solvation, entropy, and protein dynamics are not captured by a gas-phase electronic energy, so this gives one ingredient of binding affinity, not the whole thing.

Suggested incremental roadmap:

1. H₂ (this project) → validate the pipeline
2. LiH or H₂O with an active space → test scaling
3. Small non-covalent dimer (e.g., water dimer) → compute a real interaction energy
4. Add noise model + error mitigation
5. Ligand fragment + binding-site fragment from a real PDB structure

11. Troubleshooting

Symptom	Likely cause	Fix
<code>ModuleNotFoundError: qiskit_nature</code>	Package not installed in the active interpreter	<code>python3 -m pip install qiskit-nature</code> , or activate the venv
<code>externally-managed-environment</code>	System Python protected (Debian/Ubuntu)	Use a virtual environment (Section 7.2)
PySCF build fails	Unsupported Python version	Use Python 3.9 to 3.12
<code>ImportError</code> / API mismatch in <code>qiskit_nature</code>	Version drift	<code>pip list grep -i qiskit;</code> <code>pin qiskit-nature==0.7.*</code> with Qiskit 1.x
VQE energy below exact	Bug (wrong sector, geometry mismatch, nuclear repulsion counted twice)	Check that both use the same <code>problem</code> and <code>mapper</code> , and that <code>nuc</code> is added once
VQE energy equals the HF energy	Optimizer did not move	Increase <code>maxiter</code> ; check <code>ansatz.num_parameters > 0</code>
Error larger than 1.6 mHa at some distances	COBYLA local minimum / early stop	Raise <code>restarts</code> and <code>maxiter</code>
Plot window does not appear	Headless environment	The PNG is still saved; use <code>matplotlib.use("Agg")</code>

12. Glossary

Term	Meaning
Ansatz	Parameterized quantum circuit used as the trial wavefunction
Basis set	Set of functions used to expand molecular orbitals
Born-Oppenheimer	Approximation treating nuclei as fixed while solving for electrons
Chemical accuracy	Energy error ≤ 1 kcal/mol ≈ 1.6 mHa
COBYLA	Gradient-free optimizer using linear approximations
Correlation energy	$E_{\text{exact}} - E_{\text{HF}}$ within a basis
FCI	Full Configuration Interaction: exact solution within a basis
Hamiltonian	Operator whose lowest eigenvalue is the ground-state energy
Hartree (Ha)	Atomic unit of energy ≈ 27.211 eV
Hartree-Fock (HF)	Single-determinant mean-field approximation
NISQ	Noisy Intermediate-Scale Quantum devices
Parity mapping	Fermion-to-qubit encoding enabling qubit reduction
Statevector	Full vector of 2^n amplitudes, simulated exactly (no sampling noise)
UCCSD	Unitary Coupled Cluster Singles and Doubles
VQE	Variational Quantum Eigensolver

13. References

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Note on numerical values: coefficients and energies in Section 4 are representative STO-3G values for H₂ near 0.735 Å. Exact digits and sign conventions can differ slightly across library versions, but the structure of the calculation and the final energies (about −1.117 Ha for HF, about −1.137 Ha for FCI/VQE) are consistent.